U.S. Energy Information Administration

World Oil Transit Chokepoints

Last Updated: March 3, 2026 | Notes | Previous years

Overview

Table 1. Volume of crude oil and petroleum liquids transported through world chokepoints and the Cape of Good Hope, 2020–1H25

million barrels per day

Location	2020	2021	2022	2023	2024	1H25
Strait of Malacca	22.8	22.1	23.0	24.0	22.5	23.2
Strait of Hormuz	19.2	19.7	21.9	21.8	20.7	20.9
Suez Canal and SUMED Pipeline	5.4	5.2	7.3	8.8	4.8	4.9
Bab el-Mandeb	5.7	6.0	8.0	9.3	4.1	4.2
Danish Straits[a]	3.1	3.1	4.2	5.0	4.9	4.9
Turkish Straits (Dardanelles)	3.2	3.3	3.2	3.5	3.6	3.7
Panama Canal[b]	1.7	1.8	2.2	2.2	2.0	2.3
Cape of Good Hope	7.9	7.2	6.1	6.2	9.3	9.1
World maritime oil trade	**74.1**	**75.9**	**78.6**	**80.2**	**79.7**	**79.8**
World total oil supply	**94.1**	**95.8**	**100.6**	**102.6**	**103.3**	**104.4**

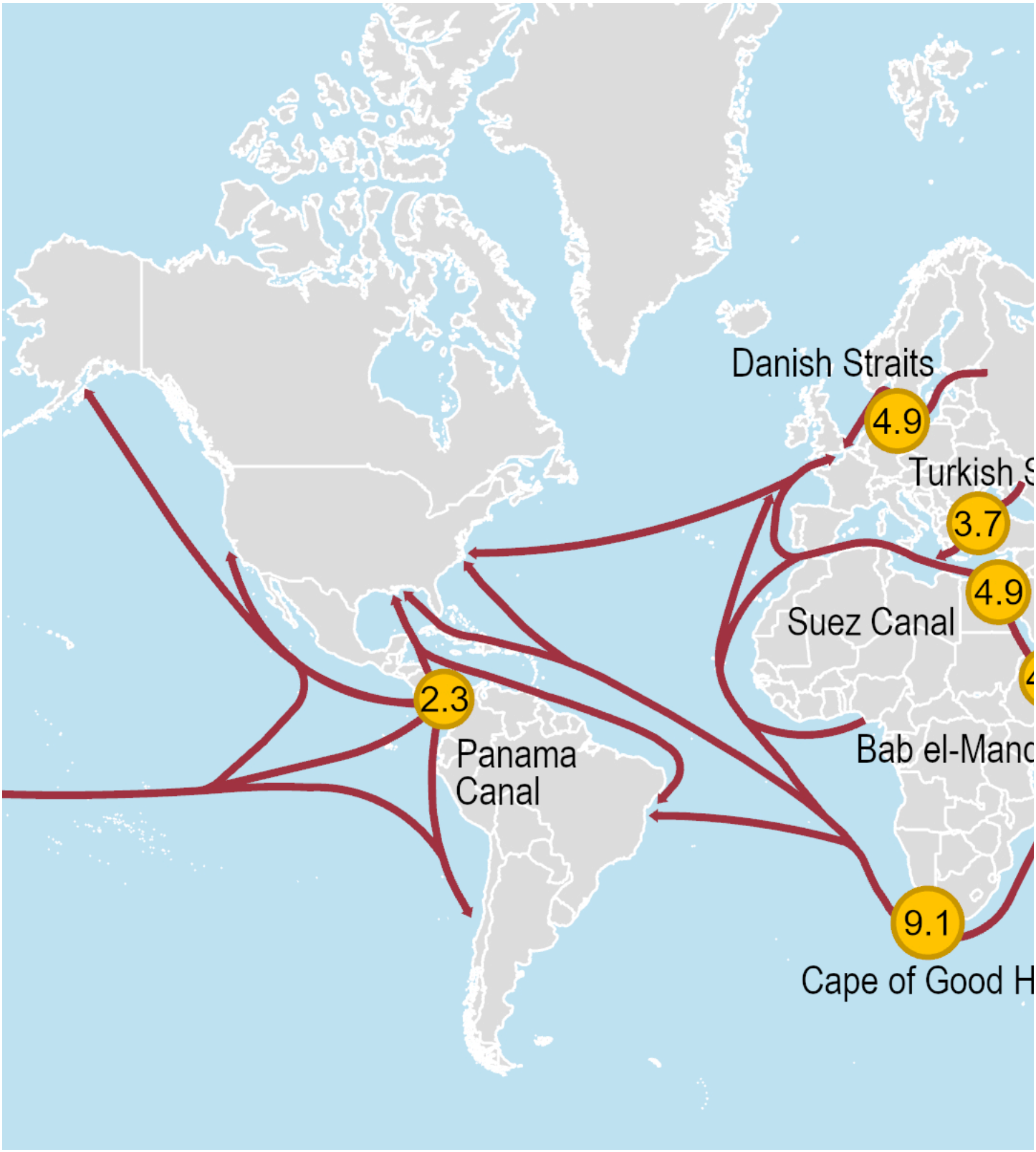
Data source: U.S. Energy Information Administration (EIA), *Short-Term Energy Outlook*, February 2026, and EIA analysis based on Vortexa tanker tracking and Panama Canal Authority data, using EIA conversion factors and calculations
Note: World maritime oil trade excludes intra-country volumes except those volumes that transit global chokepoints and the Cape of Good Hope. 1H25=first half of 2025
a The Danish Straits do not include flows through the Kiel Canal.
b Data for the Panama Canal are by fiscal year (October 1 to September 30).

Related links

- *Short-term Energy Outlook*
- Amid regional conflict, the Strait of Hormuz remains critical oil chokepoint
- About one-fifth of global liquefied natural gas trade flows through the Strait of Hormuz
- Red Sea disruptions increase oil flows around Cape of Good Hope
- Drought at the Panama Canal delays energy shipments, increasing shipping costs
- U.S. energy flows through Panama Canal rose slightly in January
- Red Sea attacks increase shipping times and freight rates
- Longer maritime routes mean more fuel consumed in shipping following Red Sea attacks

- Chokepoints are narrow channels along widely used global sea routes that are critical to global energy trade and security because of the large volumes of petroleum and other liquids and liquified natural gas that pass through them.
- International energy markets depend on reliable transport routes. The blockage of oil transit through a major chokepoint, even temporarily, can lead to substantial supply delays and higher shipping costs, resulting in higher world energy prices. Although most chokepoints can be circumvented by using other routes—which adds significantly to transit time—some chokepoints have no practical alternatives.
- This report analyzes seven chokepoints, disruptions to which could add thousands of miles of transit in alternative routes and affect oil and natural gas prices. The world's most important strategic chokepoints by volume of oil transit are the Strait of Hormuz, leading out of the Persian Gulf, and the Strait of Malacca, which links the Indian and Pacific Oceans (Figure 1). This report also discusses the Cape of Good Hope, which is not a chokepoint but is a major trade route and alternative route to other chokepoints.
- In the first half of 2025 (1H25), total world petroleum and other liquids supply was about 104.4 million barrels per day (b/d).[1] We estimate that about 76% of that amount (79.8 million b/d) traveled via seaborne trade (Table 1).[2] Oil tankers accounted for 28% of the world's shipping by deadweight tonnage in 2024, according to data from the United Nations Conference on Trade and Development (UNCTAD).[3]
- In 2024, global natural gas supply was about 407 billion cubic feet per day (Bcf/d).[4] We estimate that about 13% of that amount (53 Bcf/d) traveled via seaborne trade.[5]

Figure 1. Map of daily transit volumes of petroleum and other liquids through world maritime oil chokepoints (million barrels per day) (1H25)



Data source: U.S. Energy Information Administration (EIA) analysis, based on Vortexa tanker tracking; World Bank; and Panama Canal Authority data, using EIA conversion factors and calculations
Note: 1H25=first half of 2025

Strait of Malacca

- The Strait of Malacca, linking the Indian Ocean and the Pacific Ocean, is the shortest sea route between Middle East oil and natural gas suppliers and growing markets in East and Southeast Asia (Figure 2).
- According to the International Maritime Bureau's Piracy Reporting Centre, piracy, including attempted theft and hijackings, is a threat to tankers in the Strait of Malacca and attacks on ships increased after 2023, especially around Singapore.[6]
- This strait is the primary chokepoint in Asia and Oceania, with an estimated 23.2 million b/d of oil flow in 1H25, equivalent to 29% of total maritime oil flows, and it is the largest chokepoint in the world in terms of oil transit volume. Crude oil generally makes up slightly more than 70% of total oil flows through the strait per year, and petroleum products account for the remainder (Table 2).
- Alternative routes around the Strait of Malacca include two smaller Pacific Ocean chokepoints in the Indonesian archipelago, the Sunda Strait and the Lombok Strait (Figure 2). Flows could also travel a longer route around the entire archipelago. In addition, there is an oil pipeline that transports Middle Eastern crude oil from Myanmar to southwest China.
- Key Persian Gulf OPEC producers (Saudi Arabia, United Arab Emirates, Kuwait, and Iraq) transported nearly 60% of crude oil going through the Strait of Malacca in 1H25 (Figure 3). Flows through the strait declined in 2024 as a result of OPEC+ crude oil production cuts; however, volumes increased again in 1H25 when production targets were increased. Despite Western sanctions on its oil exports, Iran has been able to consistently increase oil exports to China since 2020. Russian volumes have fluctuated since 2022 because of sanctions and only accounted for 2% of total flows through the strait in 1H25.
- Most of the crude oil flows that traverse the Strait of Malacca are destined for East Asian countries from the Middle East (Figure 3). China accounted for 48% of the import volumes passing through the strait in 1H25. Flows to India from all Russian ports have been higher in recent years following U.S. and EU import restrictions on Russian oil, but flows through the Strait of Malacca change as volumes from Russia's Pacific coast to India vary.
- The United States exports and imports crude oil volumes that traverse the Strait of Malacca. In 1H25, the United States sent 800,000 b/d of crude oil and condensates from its Atlantic coast through Malacca to East Asia, and the U.S. Pacific coast received 200,000 b/d of cargoes via Malacca primarily from the Middle East.
- In 1H25, approximately 9.2 billion cubic feet per day (Bcf/d) of liquefied natural gas (LNG) flowed through the Strait of Malacca (Table 2). Flows from Qatar to China more than doubled since 2020, growing from 14% of total LNG flows through the Strait of Malacca in 2020 to 28% in 1H25.

Figure 2. Map of Strait of Malacca and Pacific Ocean chokepoints



Data source: U.S. Energy Information Administration

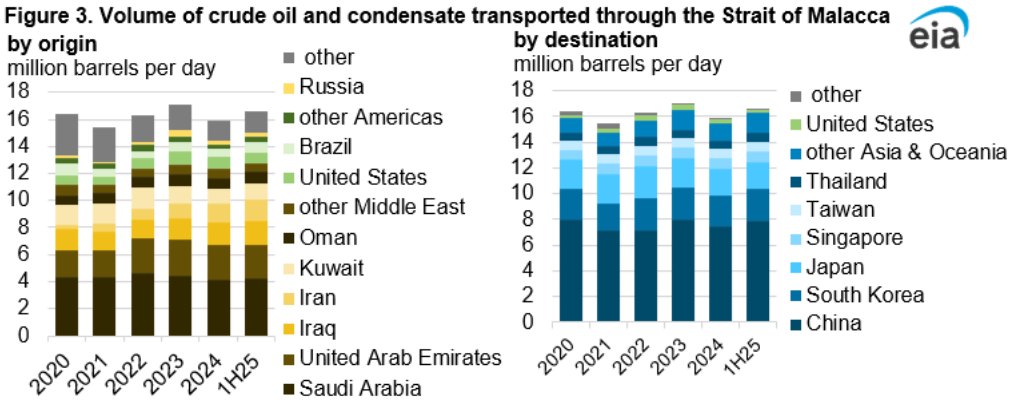
Table 2. Volume of crude oil, condensate, petroleum products, and liquefied natural gas transported through the Strait of Malacca, 2020–1H25

million barrels per day

	2020	2021	2022	2023	2024	1H25
Total oil flows through the Strait of Malacca	**22.8**	**22.1**	**23.0**	**24.0**	**22.5**	**23.2**
Crude oil and condensate	16.4	15.4	16.3	17.1	15.9	16.6
Petroleum products	6.4	6.6	6.7	6.9	6.6	6.5
LNG flows through Strait of Malacca (billion cubic feet per day)	**7.4**	**8.9**	**8.3**	**9.4**	**10.1**	**9.2**

Data source: U.S. Energy Information Administration analysis based on Vortexa tanker tracking data
Note: LNG=liquefied natural gas; 1H25=first half of 2025

Figure 3. Volume of crude oil and condensate transported through the Strait of Malacca



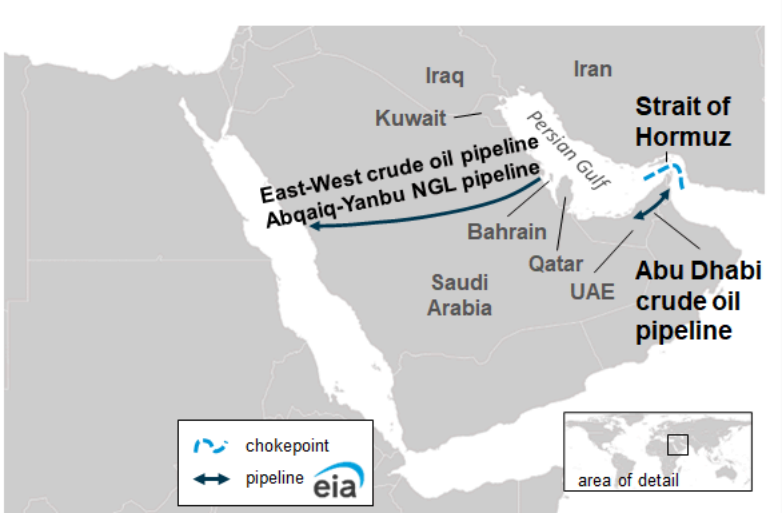
Data source: U.S. Energy Information Administration analysis based on Vortexa
Notes: 1H25= first half 2025. Dates are based on loading date.

figure data

Strait of Hormuz

- The Strait of Hormuz, located between Oman and Iran, connects the Persian Gulf with the Gulf of Oman and the Arabian Sea (Figure 4). The Strait of Hormuz is deep enough and wide enough to handle the world's largest crude oil tankers, and it is one of the world's most important oil chokepoints. Large volumes of oil flow through the strait, and, if it were to be closed, alternatives that exist could move only a portion of the oil volumes out of the strait.
- In 1H25, total oil flows through the strait averaged 20.9 million b/d, or the equivalent of about 20% of global petroleum liquids consumption, and one-quarter of total global maritime traded oil.[7] Between 2022 and 1H25, volumes of crude oil and condensate transiting the Strait of Hormuz declined by 1.5 million b/d, only partially offset by a 0.5-million b/d increase in petroleum product cargoes (Table 3 and Figure 5).[8]
- Pipelines in Saudi Arabia, the UAE, and Iran provide alternatives to the Strait of Hormuz (Figure 4). Saudi Aramco's East-West crude oil pipeline and the UAE's Abu Dhabi pipeline together could provide about 4.7 million b/d of capacity to bypass the strait in the event of a supply disruption.[9] By 2027, the UAE plans to build another 1.5 million b/d pipeline circumventing the strait, from the Jebel Dhanna export terminal to Fujairah.[10] Iran inaugurated the Goreh-Jask oil pipeline and the Jask oil export terminal on the Gulf of Oman with a single export cargo in 2021, and it sent a few small loadings in late 2024. The pipeline's effective capacity remains around 0.3 million b/d.[11]
- Declining crude oil transit through the Strait of Hormuz after 2022 reflects several trends: voluntary cuts by OPEC+ member countries starting in 2023; oil disruptions around the Bab el-Mandeb Strait, which shifted some oil flows from Saudi Arabia to Europe away from Hormuz (and subsequently the Bab el-Mandeb) to its East-West Pipeline; and higher refining capacity in the Persian Gulf states. The OPEC+ cuts from Persian Gulf states offset the higher crude oil exports from Iran, which is not subject to OPEC+ production cuts, most of which were destined for China.
- Saudi Arabia moves more crude oil and condensate through the Strait of Hormuz than any other country. Around 0.3 million b/d of intra-Saudi Arabia volumes transited the strait in 2023 from Saudi ports in the Persian Gulf, most notably Ras Tanura, to Saudi ports in the Red Sea. Red Sea attacks and oil disruptions in the Bab el-Mandeb Strait that began at the end of 2023 caused these intra-Saudi Arabia volumes to drop to 18,000 b/d by 1H25.
- We estimate that 89% of the crude oil and condensate that moved through the Strait of Hormuz went to Asian markets in 1H25. China, India, Japan, and South Korea were the top destinations, accounting for a combined 74% of all Hormuz crude oil and condensate flows in 1H25.
- In 1H25, the United States imported about 0.4 million b/d of crude oil and condensate from Persian Gulf countries through the Strait of Hormuz, accounting for about 7% of U.S. crude oil and condensate imports and 2% of U.S. petroleum liquids consumption.[12] U.S. crude oil imports from countries in the Persian Gulf have fallen to the lowest level in 40 years as domestic production has increased.
- In 1H25, 11.4 Bcf/d, or over 20% of global LNG trade, transited the Strait of Hormuz, primarily from Qatar (Table 3). China was the largest destination for LNG flows transiting Hormuz, with almost one-third of total LNG volumes destinated for China in 1H25 passing through Hormuz.

Figure 4. Map of the Strait of Hormuz and the Arabian Peninsula



Data source: U.S. Energy Information Administration
Note: NGL=natural gas liquids

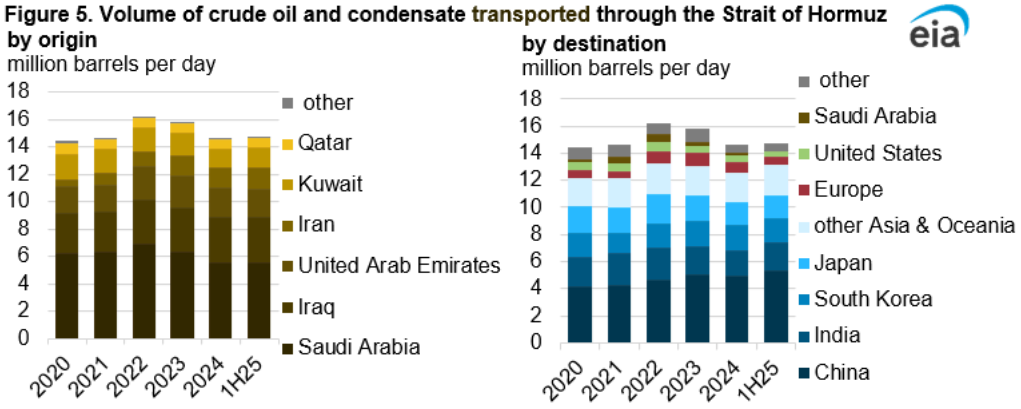
Table 3. Volume of crude oil, condensate, and petroleum products transported through the Strait of Hormuz, 2020–1H25

million barrels per day

	2020	2021	2022	2023	2024	1H25
Total oil flows through the Strait of Hormuz	**19.2**	**19.7**	**21.9**	**21.8**	**20.7**	**20.9**
Crude oil and condensate	14.4	14.7	16.2	15.8	14.6	14.7
Petroleum products	4.8	5.1	5.6	6.0	6.1	6.1
LNG flows through Strait of Hormuz (billion cubic feet per day)	**10.7**	**10.8**	**11.0**	**10.6**	**10.5**	**11.4**

Data source: U.S. Energy Information Administration analysis based on Vortexa tanker tracking data
Note: LNG=liquefied natural gas; 1H25=first half of 2025

Figure 5. Volume of crude oil and condensate transported through the Strait of Hormuz



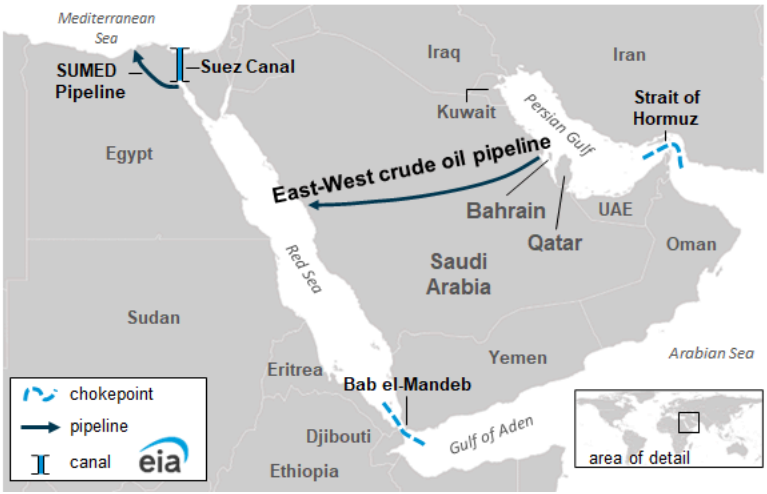
Data source: U.S. Energy Information Administration analysis based on Vortexa
Notes: 1H25= first half 2025. Dates are based on loading date.

figure data

Suez Canal, SUMED pipeline, and Bab el-Mandeb Strait

- The Suez Canal, the SUMED pipeline, and the Bab el-Mandeb Strait are strategic routes for Persian Gulf oil and natural gas shipments to Europe (Figure 6). Total oil shipments via these routes accounted for about 6% of total seaborne-traded oil in 1H25.[13]
- The Suez Canal and SUMED pipeline are in Egypt and connect the Red Sea with the Mediterranean Sea. The SUMED pipeline, owned by Mubadala Energy, transports crude oil north through Egypt between the ports of Ain Sukhna and Sidi Kerir and has a capacity of 2.5 million b/d.[14]
- The Bab el-Mandeb Strait is between the Horn of Africa and the Middle East, connecting the Red Sea to the Gulf of Aden and Arabian Sea.
- Prior to 2024, most exports of petroleum and natural gas from the Persian Gulf to Europe and the Atlantic coast of North America passed through the Suez Canal or the SUMED pipeline and both the Bab el-Mandeb and the Strait of Hormuz. However, after Yemen-based Houthi militia attacks on commercial ships transiting the Red Sea started in November 2023, some vessels began taking longer, more costly routes around the Cape of Good Hope, which avoids the Bab el-Mandeb as well as the Suez Canal.
- About 4.9 million b/d of crude oil and petroleum products transited the Suez Canal and the SUMED pipeline in 1H25, and an estimated 4.2 million b/d also transited the Bab el-Mandeb. Both volumes were approximately half of the flows in 2023 (Table 4).
- Saudi crude oil and condensate volumes flowing via the Bab el-Mandeb also declined by more than 50% from 2023 to 2024; however, Saudi volumes transiting Suez and SUMED declined only 10% as Saudi Arabia increased exports via the East-West crude oil pipeline and Red Sea terminals.
- In 1H25, Russia moved more crude oil and condensate through the Suez Canal and the Bab el-Mandeb than any other country. After the war in Ukraine began in February 2022, most of Russia's exports from the west of the country shifted from Europe to Asia, mainly India. Russian ships have rarely been targeted by Houthi militia attacks in the Bab el-Mandeb, and volumes of Russian crude oil and condensate passing through the Suez Canal and the Bab el-Mandeb in 2024 and 1H25 were little changed from 2023 (Figure 7).
- LNG flows via the Bab el-Mandeb were near zero in 2024 and 1H25 (Table 4). LNG flows through both the Suez Canal and the Bab el-Mandeb declined sharply in 2024 as ships avoided the Bab el-Mandeb because of security concerns and high insurance rates. Since 2023, LNG flows through the Suez Canal are almost entirely for delivery to Egypt or Jordan.

Figure 6. Map of Arabian Peninsula maritime chokepoints



Data source: U.S. Energy Information Administration

Table 4. Volume of crude oil, condensate, and petroleum products transported through the Suez Canal, SUMED pipeline, and Bab el-Mandeb Strait, 2020–1H25

million barrels per day

	2020	2021	2022	2023	2024	1H25
Total oil flows through the Suez Canal and SUMED pipeline	**5.4**	**5.2**	**7.3**	**8.8**	**4.8**	**4.9**
Crude oil and condensate	2.6	2.2	3.6	4.5	3.0	2.8
Petroleum products	2.7	3.0	3.7	4.3	1.7	2.1
LNG flows through the Suez Canal (billion cubic feet per day)	**3.8**	**4.7**	**4.6**	**4.1**	**0.5**	**0.9**
Total oil flows through Bab el-Mandeb Strait	**5.7**	**6.0**	**8.0**	**9.3**	**4.1**	**4.2**
Crude oil and condensate	2.7	2.7	4.0	4.7	2.7	2.4

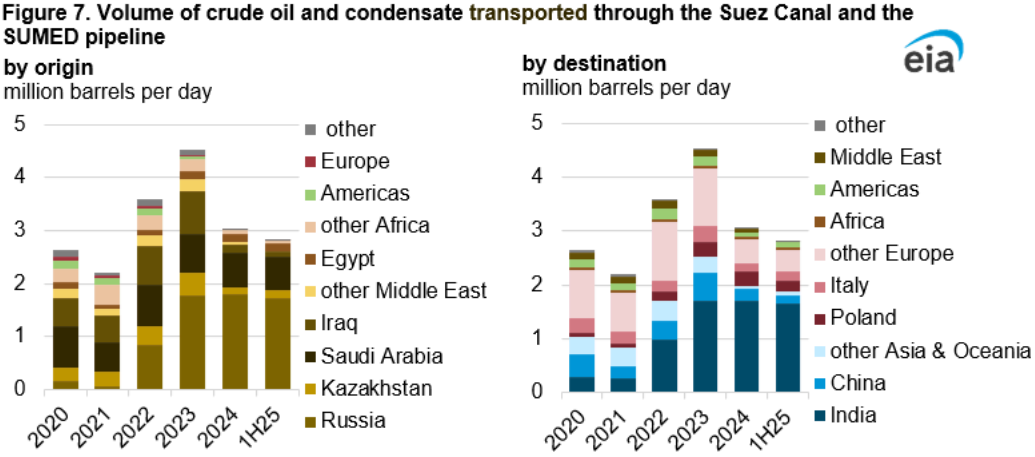
Data source: U.S. Energy Information Administration analysis based on Vortexa tanker tracking data
Note: LNG=liquefied natural gas; 1H25=first half of 2025

million barrels per day

	2020	2021	2022	2023	2024	1H25
Petroleum products	3.0	3.4	4.0	4.6	1.4	1.8
LNG flows through Bab el-Mandeb Strait (billion cubic feet per day)	**3.8**	**4.7**	**4.6**	**4.2**	**0.0**	**0.0**

Data source: U.S. Energy Information Administration analysis based on Vortexa tanker tracking data
Note: LNG=liquefied natural gas; 1H25=first half of 2025

Figure 7. Volume of crude oil and condensate transported through the Suez Canal and the SUMED pipeline



Data source: U.S. Energy Information Administration analysis based on Vortexa
Notes: 1H25= first half 2025. Dates are based on loading date.

figure data

Danish Straits

- The Danish Straits are a series of channels that connect the Baltic Sea to the North Sea (Figure 8).
- Historically, the Danish Straits were an important route for Russia's seaborne oil exports to Europe before global trade patterns shifted following the start of the war in Ukraine in 2022 and subsequent sanctions on Russia's oil exports by the EU.[15]
- An estimated 4.9 million b/d of crude oil and petroleum products flowed through the Danish Straits in 1H25, equivalent to 6% of global maritime trade. This volume is almost 60% higher than in 2021 because of the shift in trade flow from Russia (Table 5).[16]
- The Kiel Canal in northern Germany provides an alternative route for oil to the Danish Straits; however, it serves small tankers, and virtually only oil products are shipped through it (Figure 8). Oil flows through the canal were nearly 200,000 b/d in 1H25. Russia used to be the largest source of oil products flowing through the canal, but its share drastically fell after 2022. At the same time, the Netherlands, Belgium, and Denmark began to transport more oil products through the canal.[17]
- Although Russia remained the largest oil exporter through the Danish Straits in 1H25, it shipped most of these volumes from the Baltic Sea to Asia and Türkiye, instead of to western Europe and the United States, representing a shift from the historical pattern. Russia's oil exports to India through the Danish Straits rose rapidly in 2023 as India took advantage of the lower prices offered by Russia compared with the Middle East and other producers (Figure 9).[18]
- Starting in 2022, oil flows crossing the western side of the Danish Straits from countries such as the United States, Norway, the United Kingdom, and Egypt to countries east of the straits, such as Poland and Finland, rose and replaced oil imports from Russia to these countries.
- In 1H25, 1.6 Bcf/d of LNG transited the Danish Straits, about triple the volume in 2020 (Table 5). The United States became the largest supplier to countries east of the Danish Straits in 2021 as U.S. LNG export infrastructure grew rapidly during the past few years.[19] Much of these volumes replaced Europe's pipeline imports from Russia starting in 2022.[20]

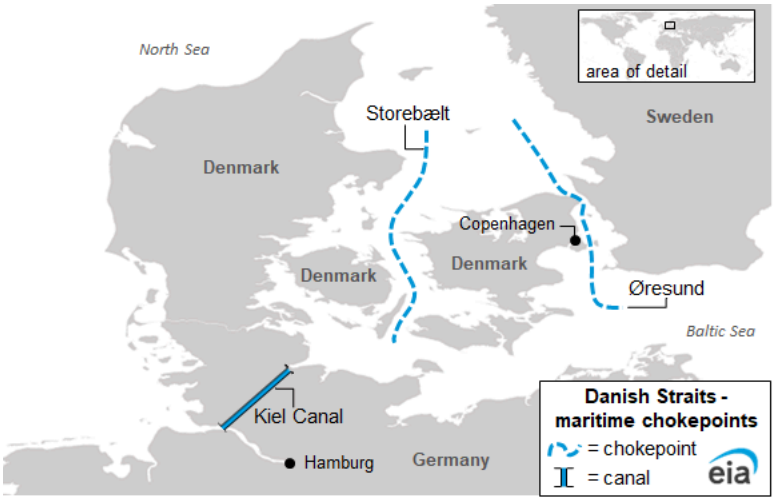


Figure 8. Map of the Danish Straits

Data source: U.S. Energy Information Administration

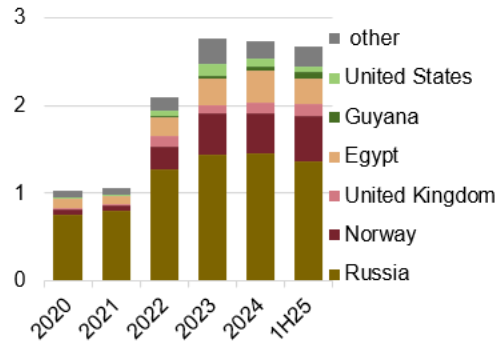
Table 5. Volume of crude oil, condensate, and petroleum products transported through the Danish Straits, 2020–1H25

million barrels per day

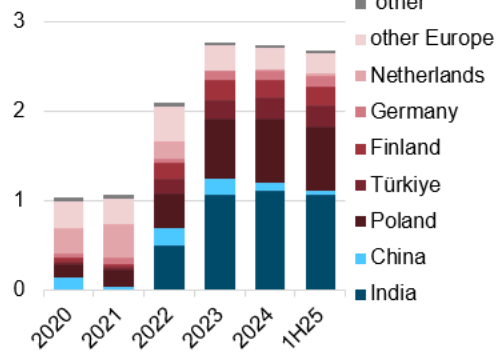
	2020	2021	2022	2023	2024	1H25
Total oil flows through the Danish Straits	**3.1**	**3.1**	**4.2**	**5.0**	**4.9**	**4.9**
Crude oil and condensate	1.0	1.1	2.1	2.8	2.7	2.7
Petroleum products	2.1	2.1	2.2	2.2	2.1	2.2
Total oil flows through the Kiel Canal	**0.2**	**0.2**	**0.2**	**0.2**	**0.2**	**0.2**
LNG flows through the Danish Straits (billion cubic feet per day)	**0.5**	**0.5**	**1.1**	**1.4**	**1.4**	**1.6**

Data source: U.S. Energy Information Administration analysis based on Vortexa tanker tracking data
Note: LNG=liquefied natural gas; 1H25=first half of 2025

Figure 9. Volume of crude oil and condensate transported through the Danish Straits
by origin
million barrels per day



by destination
million barrels per day



Data source: Vortexa.
Notes: 1H25= first half 2025. Dates are based on loading date.

figure data

Turkish Straits

- The Turkish Straits, which includes the Bosporus and Dardanelles waterways (also known as the Istanbul Strait and the Çanakkale Strait, respectively), divide Asia from Europe (Figure 10). The Bosporus is a 17-nautical mile waterway that connects the Black Sea with the Sea of Marmara. The Dardanelles is a 37-nautical mile waterway that links the Sea of Marmara with the Aegean and Mediterranean Seas. Both waterways are in Türkiye and supply Asia, western Europe, and southern Europe with oil from Russia and the Caspian Sea region.[21]
- Less than half a nautical mile wide at the narrowest point, the Turkish Straits are among the world's most difficult waterways to navigate. More than 45,000 vessels transited the straits in 2024, making the Turkish Straits one of the world's busiest maritime chokepoints.[22]
- An estimated 3.7 million b/d of crude oil and petroleum products flowed through the Turkish Straits in 1H25, equivalent to 5% of global maritime trade. About 60% of this volume was crude oil, and the remainder was oil products (Table 6).[23]
- Black Sea ports are one of the primary export routes for both crude oil and oil products from Russia and other Eurasian countries, including Azerbaijan and Kazakhstan (Figure 11). Kazakhstan, the largest exporter through the Turkish Straits, and Russia increased crude oil and condensate sent through the straits by 400,000 b/d between 2022 and 1H25. Kazakhstan exported more crude oil to Europe through the Caspian Pipeline Consortium oil pipeline using Russia's Black Sea port of Novorossiysk, and Russia sent more crude oil to India.
- Alternatives to transiting the Turkish Straits include the Baku-Tbilisi-Ceyhan pipeline from Azerbaijan in the Caspian Sea to the port of Ceyhan in Türkiye, and the Iraq-to-Türkiye pipeline from the Kurdistan region in northern Iraq, which returned to operations in September 2025 after being out of service since March 2023.[24]
- Türkiye does not allow LNG cargoes to pass through the Bosporus Strait because of safety concerns but does allow cargoes through the Strait of Dardanelles.[25] In 1H25, Türkiye imported around 0.6 Bcf/d of natural gas through its Marmara LNG regasification terminal, mostly from the United States and Algeria (Table 6).[26] The terminal is located on the Sea of Marmara inside the Strait of Dardanelles, so LNG carriers travel eastbound through the strait to reach it.
- If a disruption occurred at the Strait of Dardanelles, Türkiye could still receive natural gas through its other LNG import terminals, which are located outside of this strait, or import natural gas via pipeline from Azerbaijan, Russia, or Iran.[27]

Figure 10. Map of Turkish Straits chokepoints



Data source: U.S. Energy Information Administration

Table 6. Volume of crude oil, condensate, and petroleum products transported through the Turkish Straits, 2020–1H25

million barrels per day

	2020	2021	2022	2023	2024	1H25
Total oil flows through the Bosporus Strait	**2.9**	**3.0**	**2.9**	**3.3**	**3.3**	**3.3**
Crude oil and condensate	1.6	1.7	1.6	1.9	1.9	2.1
Petroleum products	1.3	1.3	1.2	1.4	1.4	1.3
Total oil flows through the Dardanelles Strait	**3.2**	**3.3**	**3.2**	**3.5**	**3.6**	**3.7**
Crude oil and condensate	1.7	1.8	1.7	1.9	2.0	2.2
Petroleum products	1.4	1.5	1.4	1.6	1.6	1.5

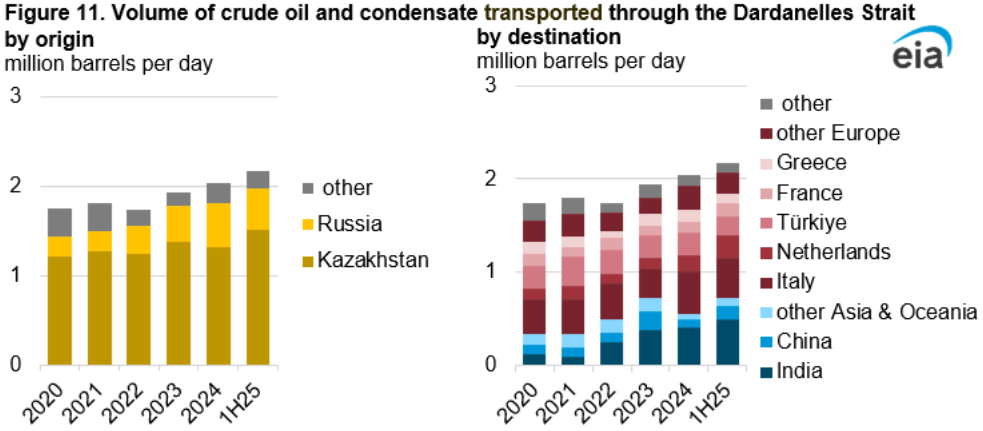
Data source: U.S. Energy Information Administration analysis based on Vortexa tanker tracking data
Note: LNG=liquefied natural gas; 1H25=first half of 2025

million barrels per day

	2020	2021	2022	2023	2024	1H25
LNG flows through the Dardanelles Strait (billion cubic feet per day)	**0.6**	**0.6**	**0.6**	**0.5**	**0.5**	**0.6**

Data source: U.S. Energy Information Administration analysis based on Vortexa tanker tracking data
Note: LNG=liquefied natural gas; 1H25=first half of 2025

Figure 11. Volume of crude oil and condensate transported through the Dardanelles Strait



Data source: Vortexa.
Notes: 1H25= first half 2025. Dates are based on loading date.

figure data

Panama Canal

- The Panama Canal connects the Pacific Ocean with the Caribbean Sea and the Atlantic Ocean. The canal is 50 miles long, and its narrowest point is the Culebra Cut (nearly 9 miles long) at the Continental Divide (Figures 12 and 13).[28]
- More than 2.3 million b/d of petroleum and other liquids were transported through the canal in FY 2025 (October 1, 2024–September 30, 2025), most of which (about 2.2 million b/d) were refined oil products (Table 7).[29]
- Although only 3% of total global maritime petroleum and petroleum product flows went through the Panama Canal in FY2025, the waterway is an important route for petroleum products—which, unlike crude oil, travel on smaller vessels able to navigate the narrow canal—and LNG traveling from the U.S. Gulf Coast to western South America and to Asia.
- Alternatives to the Panama Canal include the Straits of Magellan, Cape Horn, and Drake Passage at the southern tip of South America, but these routes add up to 8,000 miles of travel or involve traversing dangerous waterways.[30] Other routes include going eastward around the Cape of Good Hope in South Africa or through the Suez Canal. The Trans-Panama Pipeline is another alternative route for crude oil transportation. It is located near the border with Costa Rica and runs from the Port of Chiriqui Grande in Bocas del Toro, Panama, on the Caribbean Sea to the Port of Charco Azul on the Pacific coast. The pipeline has a capacity of 864,000 b/d of crude oil,[31] and the pipeline transported nearly 400,000 b/d in 2024 and 2025.[32]
- Exports of hydrocarbon gas liquids (HGLs), especially propane, from the United States to Asia have increased significantly since 2012 as U.S. production of HGLs rose and Asia's petrochemical sector developed.[33] The shortest and preferred route for tankers carrying HGLs to East Asia from the U.S. Gulf Coast is through the Panama Canal.[34] China's tariff on propane imports from the United States, imposed in the second quarter of 2025, caused a drop in propane volumes transiting the Panama Canal.[35] U.S. exports to China of ethane, another HGL, averaged more than 230,000 b/d in 2024[36] but dropped in the second quarter of 2025 before the two countries resolved tariff and licensing questions.[37]
- Insufficient water levels at Gatún Lake, which supplies water to operate the canal lock, have disrupted flows through the Panama Canal in the past (Figure 13). Total oil flows through the canal fell in FY 2024 because of drought, limiting ship traffic.[38] Constraints at the Panama Canal led to long delays, higher freight rates for vessels carrying liquefied petroleum gas, and rerouting of cargoes through the Suez Canal or around the Cape of Good Hope.[39] Oil product flows through the canal recovered in FY 2025 after water levels returned to average levels.
- In FY 2025, LNG cargoes through the Panama Canal declined to less than 0.3 Bcf/d from around 2.5 Bcf/d in FY 2021, as lower volumes of U.S. LNG transited the canal because of declining water levels and rerouting of volumes around the Cape of Good Hope. U.S. LNG transiting the Panama Canal has remained low in FY 2025, despite higher water levels. [40]

Figure 12. Map of Panama



Data source: Central Intelligence Agency, World Factbook

Figure 13. Panama Canal and Lock System, as of February 2026



Data source: U.S. Energy Information Administration

Table 7. Panama Canal oil flows, 2020–2025 (fiscal years)

million barrels per day

	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024	FY 2025
Panama Canal total oil	**1.7**	**1.8**	**2.2**	**2.2**	**2.0**	**2.3**
Crude oil	0.2	0.1	0.1	0.1	0.1	0.1
Refined products	1.6	1.7	2.0	2.2	2.0	2.2

Data source: Panama Canal Authority (with EIA conversions and calculations); Vortexa tanker tracking data[41]
Note: Totals may not equal the sum of the components due to independent rounding. Data for the Panama Canal are by fiscal year (October 1 to September 30).

million barrels per day

	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024	FY 2025
Panama Canal southbound flows	**1.6**	**1.7**	**2.0**	**2.1**	**1.9**	**2.2**
Crude oil	0.1	0.1	0.1	0.1	<0.1	0.1
Refined products	1.4	1.6	1.9	2.0	1.9	2.1
Panama Canal northbound flows	**0.2**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**
Crude oil	<0.1	<0.1	<0.1	<0.1	<0.1	<0.1
Refined products	0.1	0.1	0.1	0.1	0.1	0.1
Trans-Panama Pipeline (calendar year)	0.3	0.3	0.3	0.3	0.4	0.4

Data source: Panama Canal Authority (with EIA conversions and calculations); Vortexa tanker tracking data[41]
Note: Totals may not equal the sum of the components due to independent rounding. Data for the Panama Canal are by fiscal year (October 1 to September 30).

Cape of Good Hope

- Although not a chokepoint, the Cape of Good Hope, located on the southern tip of South Africa, is a major global trade route and a significant transit point for oil tanker and LNG shipments around the world.
- We estimate about 9.1 million b/d of seaborne-traded crude oil and petroleum products, or 11% of all seaborne-traded oil, went around the cape in both directions in 1H25, an increase of 3 million b/d from 2022 (Table 8).
- The cape is an alternative sea route for vessels traveling westward that want to bypass the Gulf of Aden, Bab el-Mandeb Strait, and the Suez Canal. However, diverting vessels around the cape significantly increases costs and shipping time. For example, closure of the Suez Canal and the SUMED Pipeline would require oil tankers to divert around the cape, adding approximately 15 days to transit from the Arabian Sea to Europe.[42]
- Oil and LNG flows to China increased after China lifted COVID pandemic lockdowns. Drought and restrictions at the Panama Canal during 2023 and disruptions at the Bab el-Mandeb Strait and Red Sea starting in November 2023 resulted in many oil and natural gas vessels taking the longer route around the cape to avoid these chokepoints. Although the disruption at the Panama Canal resolved in 2024, disruptions at the Suez Canal and the Bab el-Mandeb continue causing energy flows around the cape to be higher than historical averages.
- Most crude oil volumes traveling around the cape move eastbound to Asian markets (5.2 million b/d) in 1H25. More than 40% of the crude oil imports traversing the cape went to China. Most of the crude oil flows going around the cape originated from the United States (23%), West Africa (28%), and South America and Mexico (29%) (Figure 14).[43]
- LNG flows around the cape reached a high of 7.8 Bcf/d in 2024, before falling again by 1H25 due to U.S.-China trade tensions and a mild winter in Asia, which reduced LNG imports to much of the region.

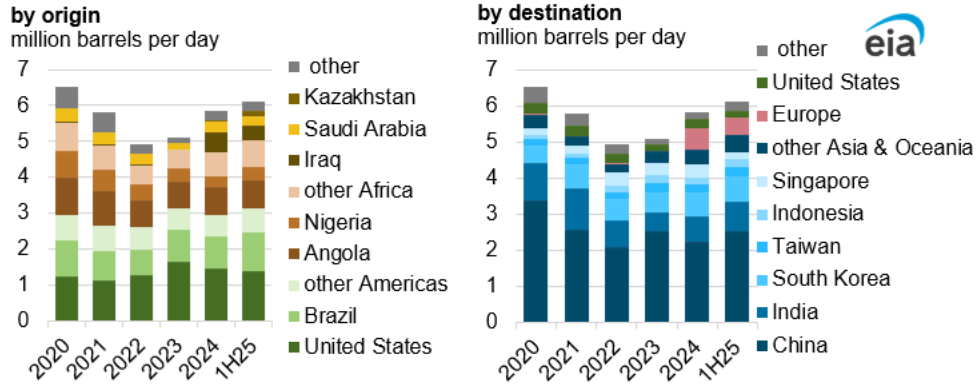
Table 8. Volume of crude oil, condensate, and petroleum products transported around the Cape of Good Hope, 2020–1H25

million barrels per day

	2020	2021	2022	2023	2024	1H25
Total oil flows around the cape	**7.9**	**7.2**	**6.1**	**6.2**	**9.3**	**9.1**
Crude oil and condensate	6.5	5.8	4.9	5.1	5.8	6.1
Petroleum products	1.3	1.4	1.1	1.1	3.4	2.9
LNG flows around the cape (billion cubic feet per day)	**3.5**	**3.3**	**1.5**	**2.1**	**7.8**	**5.7**

Data source: U.S. Energy Information Administration analysis based on Vortexa tanker tracking data
Note: LNG=liquefied natural gas; 1H25=first half of 2025

Figure 14. Volume of crude oil and condensate transported around the Cape of Good Hope



Data source: Vortexa
Notes: 1H25= first half 2025. Dates are based on loading date.

figure data

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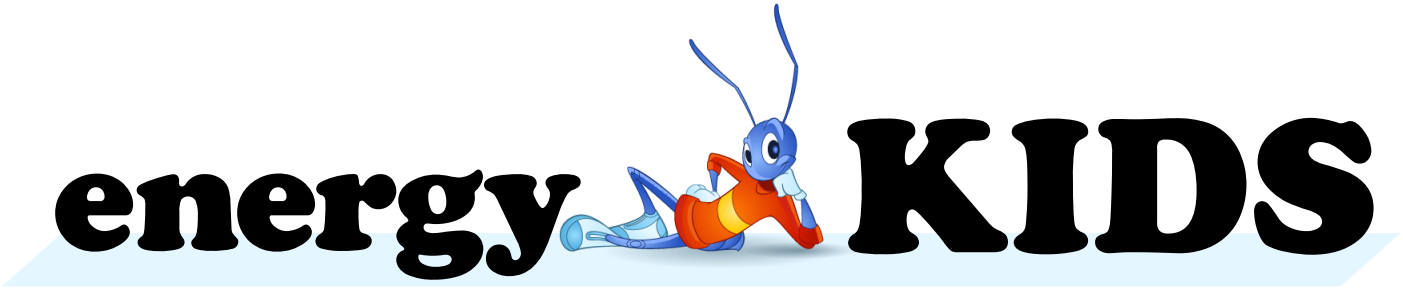
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